Kevin Kuister

Zusammenfassung: Hitchcock-Transportproblem

Hitchcock-Transportproblem

Transportproblem

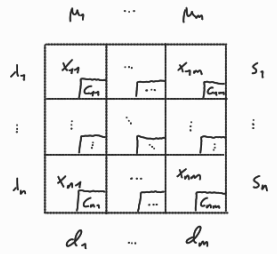
- Lieferanten $i = 1, \dots, n$
- Produzierte Einheiten s_i eines Produkts
- Verbraucher $j = 1, \dots, m$
- Bedarf d_j
- $\sum_{i=1}^{n} s_i = \sum_{j=1}^{m} d_j$
- Transportkosten $i \to j$: c_{ij}
- Ziel: Kostenminimierung
  - $\hookrightarrow \min \sum_{i=1}^{n} \sum_{j=1}^{m} c_{ij} \cdot x_{ij}$ unter

$$\sum_{i=1}^{n} x_{ij} = d_j \ \forall j \ \wedge \ \sum_{j=1}^{m} x_{ij} = s_i \ \forall i \ \wedge \ x_{ij} \geq 0 \ \forall i,j$$

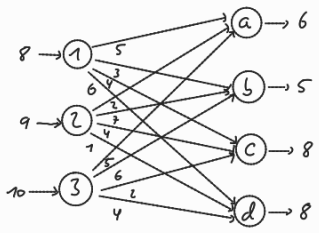
Tableau-Form

λ_i $i = 1, \dots, n$ } duale Variablen

μ_j $j = 1, \dots, m$ }



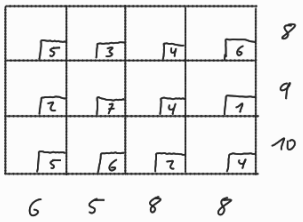
Beispiel



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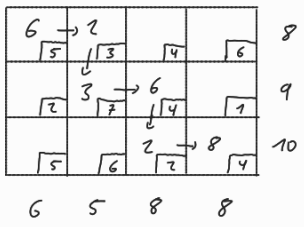
Quelle \ Senke	1	2	3	4	Angebot
1	5	3	4	6	8
2	2	7	4	1	9
3	5	6	2	4	10
Nachfrage	6	5	8	8	27

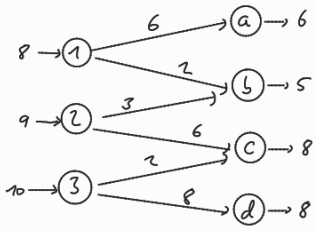
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1.) Basislösung

- Horizontal durch Lieferanten/Verbrauch-Paare (i,j)
- Erhöhen von x_{ij} bis s_i oder d_j erfüllt
  - nächste Zeile
  - nächste Spalte



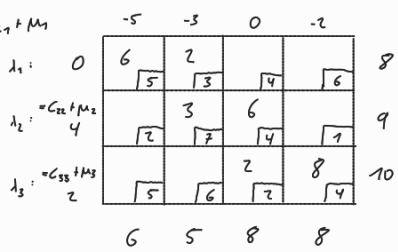


2.) Duale Variablen

$c_{ij} = \lambda_i - \mu_j$, für (i,j) Basislösung, $\lambda_1 = 0$

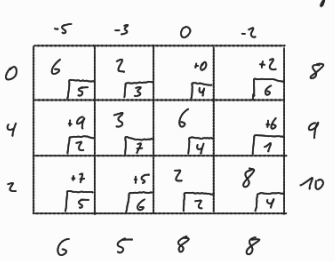
$c_{21} = \lambda_2 - \mu_1$ $\quad \mu_1 = \lambda_1 - c_{11}$, $\mu_2 = \lambda_1 - c_{12}$, $\mu_3 = \lambda_2 - c_{23}$, $\mu_4 = \lambda_3 - c_{34}$

$\lambda_2 = c_{21} + \mu_1$



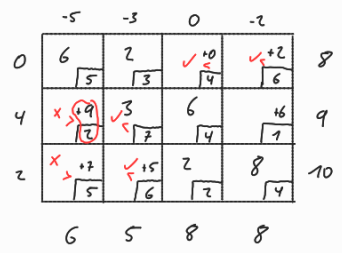
3.) Reduzierte Kosten

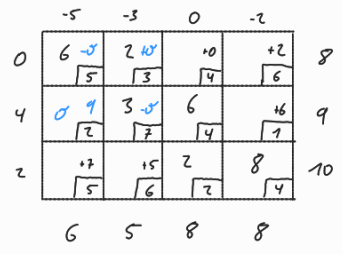
$\lambda_i - \mu_j$ für (i,j) nicht-Basislösung

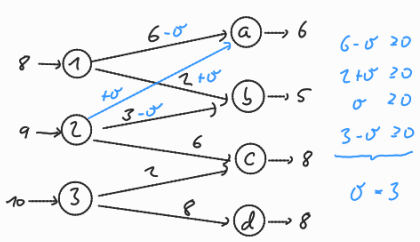


4.) Netzwerk-Simplex

Optimale Lösung: $c_{ij} \geq \lambda_i - \mu_j \ \forall (i,j)$ nicht-Basislösung







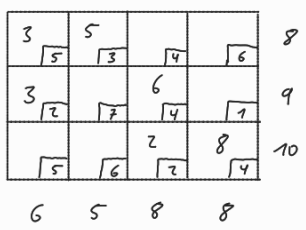
$6 - \sigma \geq 0$

$2 + \sigma \geq 0$

$\sigma \geq 0$

$3 - \sigma \geq 0$

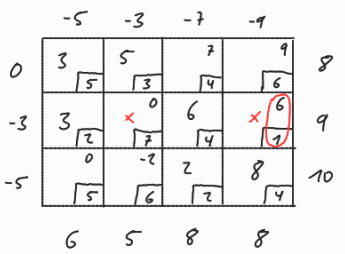
$\sigma = 3$

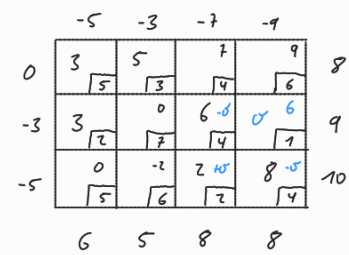


0-Zelle fällt weg;

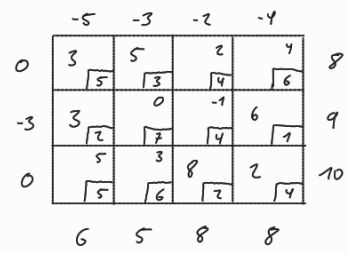
Falls ≥ 2 Nullzellen

$\hookrightarrow$ Fallunterscheidung





$\sigma = 6$



$\Rightarrow$ Optimal